

Dr. Suman Gare

Computational Biologist | Single-Cell & Spatial Bioinformatics

Email: garesuman@gmail.com | Phone: +49 15754942165

LinkedIn: <https://www.linkedin.com/in/gare-suman-phd-02b6a77a/> | GitHub: <https://github.com/sumanchemphd>

PROFESSIONAL SUMMARY

Computational Biologist and Bioinformatics Scientist with over 8 years of interdisciplinary research experience in **single-cell genomics, spatial transcriptomics, multi-omics, systems biology, and computational biology**. Experienced in developing reproducible bioinformatics workflows for high-dimensional biological datasets and translating computational results into biologically meaningful insights. Proficient in **Python, R, Bash, Linux, Git, and HPC**, with experience in scRNA-seq analysis, spatial analysis, differential expression, pathway enrichment, cell-cell communication, statistical modelling, and mechanistic systems biology. Experienced in interdisciplinary collaboration with experimental researchers and in communicating computational findings through publications and scientific presentations.

CORE COMPETENCIES

Single-cell RNA-seq • Spatial Transcriptomics • Multi-omics Integration • NGS Data Analysis • Bulk Transcriptomics • Cell-Type Annotation • Cell-State Analysis • Differential Expression • Pathway Enrichment • Cell-Cell Communication • Spatial Neighborhood Analysis • Biomarker Discovery • Statistical Analysis • Machine Learning • Reproducible Bioinformatics Workflows • Systems Biology

TECHNICAL SKILLS

Programming: Python, R, MATLAB, Bash

Platforms & Reproducibility: Linux, HPC, Git/GitHub, Conda

Bioinformatics: Seurat, Scanpy, Squidpy, CellChat, DESeq2, edgeR, limma, clusterProfiler, SingleR, AnnData

Data Types: Single-cell RNA-seq, Spatial Transcriptomics, Bulk Transcriptomics, Multi-omics

Quantitative Methods: PCA, UMAP, Clustering, HDBSCAN, Statistical Modelling, ODE/PDE, Agent-Based Modelling, Parameter Estimation

PROFESSIONAL EXPERIENCE

Postdoctoral Scientist

University Hospital Bonn, Germany | September 2023 – February 2026

- Conducted interdisciplinary computational research on **cellular signaling, spatial organization, and the tumor microenvironment**.
- Developed reproducible computational and bioinformatics workflows using **Python, R, Bash, Linux, Git, and HPC**.
- Applied single-cell and spatial transcriptomics approaches for characterization of cellular populations, molecular programs, and tissue organization.

- Performed **differential expression, pathway enrichment, cell–cell communication, statistical analysis, and biological interpretation** of high-dimensional datasets.
- Developed quantitative models of **IL-2/STAT5 signaling**, including ODE/PDE and spatial approaches to investigate signaling dynamics, cellular motility, and feedback mechanisms.
- Worked closely with experimental researchers to interpret computational results and connect model predictions with biological observations.
- Contributed to scientific publications, presentations, and interdisciplinary research discussions.

SELECTED COMPUTATIONAL BIOLOGY PROJECTS

Single-Cell Transcriptomics – Liver Disease

- Developed an end-to-end scRNA-seq workflow for approximately **66,000 cells** from cholestatic and healthy human liver samples.
- Performed QC, doublet filtering, integration, clustering, cell-type characterization, pseudobulk analysis, and pathway-level investigation.
- Investigated hepatocyte and epithelial programs related to **bile-acid synthesis, transport, and signaling**.
- Developed a modular and reproducible **R/Bash/Linux/HPC/GitHub analysis pipeline**.

Single-Cell Analysis – Tumor Microenvironment

- Analyzed approximately **21,000 high-quality NSCLC single cells** and characterized **24 cellular populations** across tumor, immune, stromal, and endothelial compartments.
- Investigated epithelial cell states and molecular programs associated with **EMT, hypoxia, proliferation, interferon response, and cellular stress**.
- Applied cell-type annotation, marker analysis, dimensionality reduction, pathway analysis, and visualization using **Seurat/Scanpy-based workflows**.

Spatial Transcriptomics – Tissue Microenvironment

- Analyzed **Visium HD spatial transcriptomics** data comprising approximately **84,000 spatially resolved cells** and >30,000 genes.
- Characterized spatial organization of neuronal, glial, vascular/endothelial, and other tissue populations.
- Applied **Scanpy and Squidpy** for spatial neighborhood analysis and identification of spatially organized multicellular tissue niches.
- Integrated gene-expression and spatial information to investigate tissue architecture and cell–cell relationships.

Single-Cell Systems Biology – MASLD/MASH

- Analyzed approximately **128,000 single cells** across healthy, fatty-liver, and NASH conditions.
- Characterized hepatocyte, **endothelial/LSEC**, Kupffer, stromal, cholangiocyte, and other cellular populations.
- Investigated disease-associated **endothelial and stromal remodeling** and intercellular communication.
- Integrated single-cell findings with statistical and mechanistic modelling for biomarker and therapeutic-target prioritization.

Multi-Omics & Regulatory Network Analysis

- Integrated mRNA and miRNA datasets to investigate disease-associated regulatory mechanisms.
- Applied differential expression, pathway enrichment, network analysis, biomarker prioritization, and statistical validation.
- Connected high-dimensional molecular data with mechanistic systems biology models to generate testable biological hypotheses.

EDUCATION

PhD – Systems Biology

Indian Institute of Technology Hyderabad, India

M.Tech – Chemical Reaction Engineering

Osmania University, India

B.Tech – Chemical Engineering

RGUKT Basar, India

SELECTED PEER-REVIEWED PUBLICATIONS

1. Gare S, Chel S, et al. (2022). *Integrated live-cell imaging and UMAP-HDBSCAN framework for calcium signaling analysis*. **Integrative Biology**, 14, 184–203.
2. Manohar K, Gare S (Equal Contribution), et al. (2021). *Quantitative confocal microscopy for calcium dose-response analysis*. **SLAS Technology**, 26, 454–467.
3. Dey A, Gare S (Equal Contribution), et al. (2022). *3D imaging of ZnO nanoparticles and ROS accumulation*. **AIChE Journal**, 68, e17801.
4. Venkateswarlu K, Gare S, et al. (2020). *3D imaging of calcium oscillations in microglial cells*. **Biotechnology and Bioengineering**, 117, 3108–3123.
5. Dhyani V, Gare S, et al. (2020). *GPCR-mediated regulation of calcium dynamics: A systems perspective*. **Cellular Signalling**, 74, 109717.
6. Manohar K, et al. (2021). *Nifedipine reduces hypoxic cell death through calcium signaling modulation*. **Free Radical Biology and Medicine**, 177, 189–200.

7. Dhyani V, Gare S, et al. (2023). *Computational modeling of neural firing heterogeneity. Hippocampus.*

Conference Publications

9 peer-reviewed international conference publications in **IEEE EMBC, IEEE ISBI, IEEE NCC, IEEE NER, and IEEE INDICON**, including 3 first-author publications.

ADDITIONAL STRENGTHS

- Strong experience working at the interface of **computational and experimental biology**
- Reproducible and well-documented computational workflow development
- Experience handling large biological datasets in **Linux/HPC environments**
- Biological interpretation of high-dimensional single-cell and spatial datasets
- Collaborative interdisciplinary research experience
- Scientific writing, visualization, presentations, and publication experience

RESEARCH INTERESTS

Single-Cell Genomics • Spatial Transcriptomics • multi-omics • Vascular Biology • Tissue Regeneration • Tumor Microenvironment • Computational Genomics • Systems Biology • Cell–Cell Communication • Therapeutic Target Discovery

REFERENCES

Available upon request.